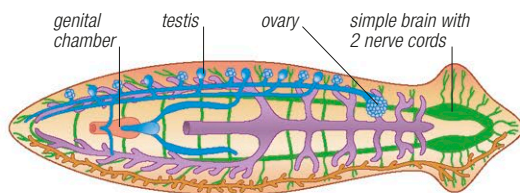


ANATOMY

All animals are made up of various parts. Their smallest fully functional parts are cells, which are shaped in different ways according to the tasks that they carry out. Groups of similar cells are organized into tissues, and tissues are grouped together to form organs. Organs themselves are linked to form organ systems, which carry out all the processes essential for survival. The structure of these systems varies widely between one type of animal and another, and also between animals that live in different ways, but the work they do is the same.

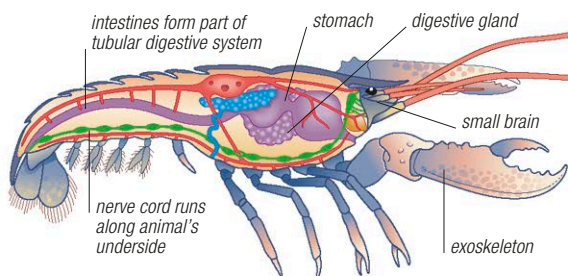
Body systems

Animals have up to a dozen separate body systems. In many species, the muscular and skeletal systems make up a large proportion of the body's total weight, while the integumentary



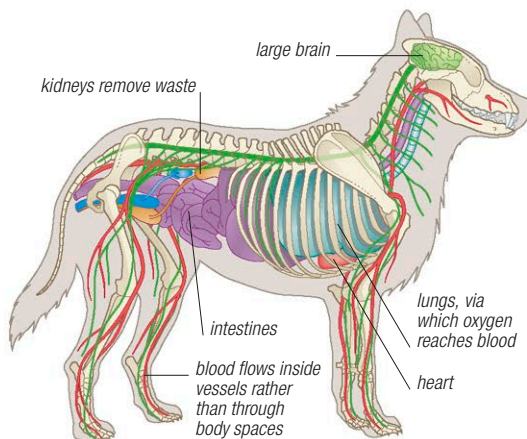
SIMPLE INVERTEBRATE BODY SYSTEMS

Simple invertebrates, such as flatworms, do not have respiratory organs or a circulatory system. The digestive system often has just one opening, the mouth, and the reproductive system is typically made up of both male and female organs.



ARTHROPOD BODY SYSTEMS

The tubular digestive system is open at both ends. The blood flows partly through vessels and partly through body spaces. Oxygen is supplied via gills or via minute airways called tracheae.



VERTEBRATE BODY SYSTEMS

The nervous system is highly developed, and the brain is typically larger than in invertebrates. The circulatory system is closed, and blood is pumped under high pressure by the heart.

KEY TO SYSTEMS

■ CIRCULATORY	■ EXCRETORY
■ DIGESTIVE	■ NERVOUS
■ RESPIRATORY	■ REPRODUCTIVE

system—the parts that form the “outer cladding”—protect the body from physical damage and, in terrestrial animals, from the threat of drying out.

Two key systems enable animals to obtain energy from food. The digestive system breaks food down so that it can be absorbed; and the respiratory system delivers oxygen to the body's cells so that food substances can be “burned” and their chemical energy released. The respiratory system also removes carbon dioxide—a potentially toxic waste product formed during the production of energy. In many animals, including all vertebrates, oxygen and carbon dioxide are carried by blood in the circulatory system. Carbon dioxide is usually exhaled; other kinds of dissolved waste are removed, before they have a chance to build up, by a separate excretory system.

Animals use 2 different systems to coordinate their bodies, and to react to their surroundings. The nervous system deals with anything that needs a fast response, processing information gathered by sense organs. In all animals, it triggers built-in or instinctive behavior, but in some, especially vertebrates, it also stores information, allowing animals to adapt their behavior according to their past experience. The endocrine system works in conjunction with the nervous system, releasing hormones, or chemical messengers, that help the nervous system coordinate long-term processes.

Finally, the reproductive system carries out the most important task: producing young. Unlike other body systems, it often functions only during a set season, and then only in mature animals.

Skeletons and support

Animals need to keep the shape of their bodies stable. Many invertebrates achieve this without any hard body parts at all. Instead, they rely on the pressure of internal fluids to keep their bodies firm in the same way that air stabilizes a tire. This system, called a hydrostatic skeleton, works well on a small scale. But in larger animals, particularly land ones, this type of skeleton is often not strong enough to support the body's weight. Animals have developed 2 quite different solutions to this problem: external shells and body cases; and internal skeletons, typically made of bone.

Shells are made up of one or 2 parts, and they grow with their owner. They can be seen in marine animals called brachiopods, or lampshells, but they are most highly developed in bivalve mollusks, which can grow shells over 3¼ ft (1 m) across. Body cases are more complex than shells: they consist of a large number of separate plates that meet at flexible joints. They are a characteristic

feature of arthropods—a huge group of invertebrates that includes insects, crustaceans, and arachnids. These cases, or exoskeletons, cover the entire body and include structures as strong as a crab's pincers or as delicate as a butterfly's antennae. Unlike shells, body cases cannot grow, so periodically they have to be shed and replaced.

Internal skeletons (endoskeletons), made of bone and/or cartilage, provide support from within. Found only in vertebrates, they have 2

great advantages: they are relatively light for their size—important for land animals that have to move quickly—and they can grow.



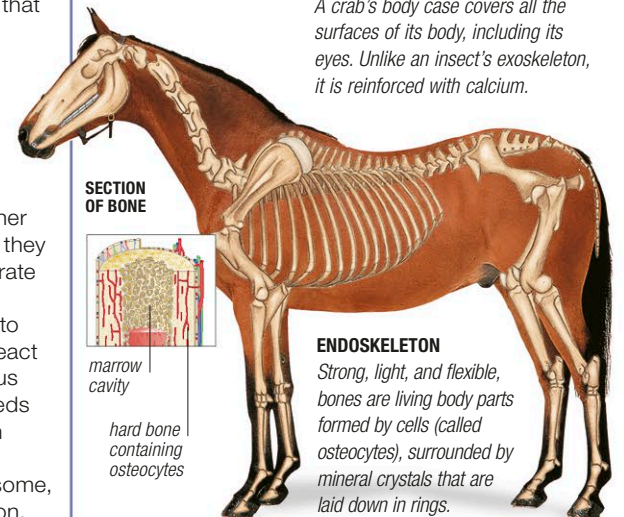
SHELL

A snail's shell grows at its lip, getting progressively larger with its owner. Some species can seal the shell with an operculum, or “door.”

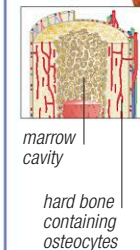


EXOSKELETON

A crab's body case covers all the surfaces of its body, including its eyes. Unlike an insect's exoskeleton, it is reinforced with calcium.



SECTION OF BONE



ENDOSKELETON

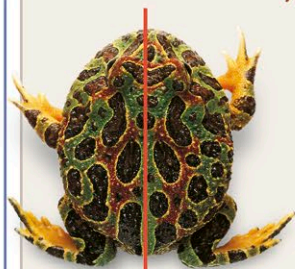
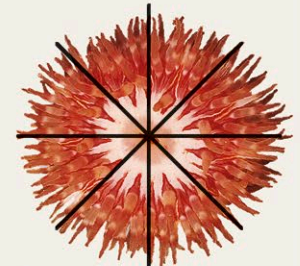
Strong, light, and flexible, bones are living body parts formed by cells (called osteocytes), surrounded by mineral crystals that are laid down in rings.

SYMMETRY

Some animals' bodies are circular, without a head or tail. This type of anatomy, called radial symmetry, can be seen in sea anemones and other cnidarians, and also in ctenophores or comb jellies. The vast majority of animals show bilateral symmetry, which means they can be divided into halves. The halves are not always equal: male fiddler crabs, for example, have very unequal claws, and flatfish have different sides.

RADIAL SYMMETRY

A sea anemone can be divided into 2 equal halves on any axis. It has a central mouth and body cavity, with feeding tentacles arranged in a ring.



BILATERAL SYMMETRY

Frogs and toads have just one axis of symmetry. Externally, the 2 halves look identical; internally, some organs lie to the left or the right.